factwise.io

# Assessment: Technical Product Manager

- This assessment has two sections and the document has 2 pages
- You are expected to submit within 24 hours of receiving this file
- Please make the submission as a PDF
- Your file should be named 'Your Name TPM' (e.g.: John Smith TPM)

**Problem Statement:** You are tasked with building an app for a virus quarantine & treatment facility From

the head doctor:

"This virus has a 15% mortality rate, and its main symptom is a prolonged fever. A patient is considered cured once they have reported 3 full days without a fever. Currently, our nurses jot down temperatures in a journal. The doctor reviews these journals when deciding what treatment to administer and informs the admin staff when they feel a patient could be discharged (solely based on the 3-day no-fever criterion).

We are understaffed and unable to keep up with the volume of patients coming in. We have rooms for up to 74 patients, which are currently at full occupancy.

Our process requires that each patient have their temperature measured by a nurse daily, and then be visited by a doctor at some point on the same day.

However, in reality, no one knows which patients have had their tempe

ratures measured, and which patients have been visited by a doctor already. So, sometimes a patient's temperature is measured multiple times on the same day, and sometimes their temperature is not measured at all. Doctor visits are also similarly erratic, and sometimes the doctor visits a patient before their temperature has been taken – as a consequence, the doctor has to waste precious minutes on doing this task themself or come back later. At times, patients end up staying an extra day or two in our centre because of these inefficiencies, or because doctors fail to promptly inform admin staff about discharges.

We want to treat as many patients as possible, so we need to be better organized, and discharge patients in a timely manner. We also want to track the quality of our treatment, so we can take measures in case the mortality rate in our centre rises above the expected 15%. Currently, we have no way of measuring our performance against the 85% success rate benchmark."

You and a small team of developers need to build an app to help this centre with its temperature tracking and administration needs. Your team needs to deliver a working product ASAP to solve the centre's basic requirements, and *nice-to-have features can be prioritized later*.

*The content of this document is confidential and intended for the recipient specified in the message only. It is strictly forbidden to share any part of this document with any third party, without a written consent of the sender. If you received this message by mistake, please reply to this message and follow with its deletion, so that we can ensure such a mistake does not occur in the future.*

**Q1) In your app's first version, what are the different kinds of users, and which features and actions are relevant for which user types?**

**Q2) Please build a functional prototype of the app using any AI tool of your choice. Please provide a deployable working link either on vercel or any platform of your choice.**

Your submission should include:

- **All basic screens** of the app (UI flow).
- **Working functionality for key actions**. For example:
  - A nurse should be able to **add and record a patient's temperature**.
  - Any other core features relevant to the app should be demonstrable.

Please ensure the prototype shows both the **screen designs** and the **actual workflow** for each feature.

**Q3) For the entire app overview, what additional logical constraints would you provide to the developers? Include details not immediately apparent from the screen designs, such as when to display warnings or errors to the user, and when to allow or disallow certain actions for specific user types, etc.**

**Q4) In the next sprint planning meeting, after reviewing the last sprint, you and your team estimate that their velocity is 25 story points. For the next sprint, you already plan to finish 3 stories of 5 points each. In addition to these, which of the following options would you add to the sprint such that the effort is minimized and the value is maximized?**

Assume that High priority features are 2X as valuable
as medium priority features, and medium priority features are 3X as valuable as low priority features.

| Feature | Priority | Effort |
|---|---|---|
| A | High | 1 |
| B | High | 3 |
| C | High | 5 |
| D | Medium | 2 |
| E | Medium | 3 |
| F | Medium | 1 |
| G | Low | 0.4 |
| H | Low | 0.5 |
| I | Low | 0.3 |
| J | Low | 0.1 |

*The content of this document is confidential and intended for the recipient specified in the message only. It is strictly forbidden to share any part of this document with any third party, without a written consent of the sender. If you received this message by mistake, please reply to this message and follow with its deletion, so that we can ensure such a mistake does not occur in the future.*




*The content of this document is confidential and intended for the recipient specified in the message only. It is strictly forbidden to share any part of this document with any third party, without a written consent of the sender. If you received this message by mistake, please reply to this message and follow with its deletion, so that we can ensure such a mistake does not occur in the future.*